

## Mode 8 — Square

DP16 (16×16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) →  
add-and-square

### Step 1 — DP16 (16×16)

$$\text{DP16 (16} \times \text{16)} = \sum_{i=0}^{15} a_i b_i$$

### Step 2 — DP8 (16×16)

$$\text{DP16 (16} \times \text{16)} = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 (16} \times \text{16)}} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 (16} \times \text{16)}}$$

### Step 3 — DP8 (16×8)

$$\text{DP16 (16} \times \text{16)} = 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}}$$

### Step 4 — DP8 (16×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} &= 2^{12} \underbrace{\sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \\ &+ 2^{12} \underbrace{\sum_{i=8}^{15} a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \end{aligned}$$

### Step 5 — DP8 (8×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} &= 2^{20} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{20} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^{12} \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^8 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

### Step 6 — DP8 (4×4)

$$\text{DP16 (16} \times \text{16)} = 2^{24} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hh}}_{\text{DP8 (4} \times \text{4)}} + 2^{20} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hh}}_{\text{DP8 (4} \times \text{4)}} + 2^{24} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hh}}_{\text{DP8 (4} \times \text{4)}} + 2^{20} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hh}}_{\text{DP8 (4} \times \text{4)}}$$

$$\begin{aligned}
& + 2^{20} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{20} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

**Step 7 — Add-and-square** (substitute each  $\text{DP8 } (4 \times 4) = \frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$ )

$$\begin{aligned}
\text{DP16 } (16 \times 16) &= 2^{24} \underbrace{\frac{1}{2} \left( \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
&+ 2^{20} \underbrace{\frac{1}{2} \left( \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
&+ 2^{24} \underbrace{\frac{1}{2} \left( \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
&+ 2^{20} \underbrace{\frac{1}{2} \left( \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
&+ 2^{20} \underbrace{\frac{1}{2} \left( \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hl})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hl})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{lh})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{lh})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hl})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hl})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hl})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hl})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{ll})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{ll})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{lh})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{lh})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{lh})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left( \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{ll})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{ll})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{ll})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$+ \frac{1}{2} \underbrace{\left( \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{ll})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{ll})^2 \right)}_{\text{DP8}(4 \times 4)}$$

Step 8 — All components

$$\begin{aligned}
\text{PE} = & 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hh})^2 + 2^{24} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{20} \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hh})^2 \\
& + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hl})^2 + 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hl})^2 + 2^{20} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hl})^2 + 2^{16} \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hl})^2 \\
& + 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hh})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hh})^2 + 2^{16} \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hh})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hh})^2 \\
& + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lh})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lh})^2 + 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{lh})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{lh})^2 \\
& + 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hl})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hl})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hl})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hl})^2 \\
& + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{ll})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{ll})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{ll})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{ll})^2 \\
& + 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lh})^2 + 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{lh})^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{lh})^2 \\
& + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{ll})^2 + \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{ll})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{ll})^2 + \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{ll})^2 \\
\alpha = & 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{24} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{20} \sum_{i=8}^{15} (a_{L,i}^{hi})^2 \\
& + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{20} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{16} \sum_{i=8}^{15} (a_{L,i}^{hi})^2 \\
& + 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^{16} \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{lo})^2 \\
& + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi})^2 \\
& + 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{lo})^2 \\
& + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi})^2 \\
& + 2^8 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo})^2 \\
& + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + \sum_{i=8}^{15} (a_{L,i}^{lo})^2 \\
\beta = & 2^{24} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{24} \sum_{i=8}^{15} (b_i^{hh})^2 + 2^{20} \sum_{i=8}^{15} (b_i^{hh})^2 \\
& + 2^{20} \sum_{i=0}^7 (b_i^{hl})^2 + 2^{16} \sum_{i=0}^7 (b_i^{hl})^2 + 2^{20} \sum_{i=8}^{15} (b_i^{hl})^2 + 2^{16} \sum_{i=8}^{15} (b_i^{hl})^2 \\
& + 2^{16} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{12} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{16} \sum_{i=8}^{15} (b_i^{hh})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{hh})^2 \\
& + 2^{16} \sum_{i=0}^7 (b_i^{lh})^2 + 2^{12} \sum_{i=0}^7 (b_i^{lh})^2 + 2^{16} \sum_{i=8}^{15} (b_i^{lh})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{lh})^2 \\
& + 2^{12} \sum_{i=0}^7 (b_i^{hl})^2 + 2^8 \sum_{i=0}^7 (b_i^{hl})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{hl})^2 + 2^8 \sum_{i=8}^{15} (b_i^{hl})^2 \\
& + 2^{12} \sum_{i=0}^7 (b_i^{ll})^2 + 2^8 \sum_{i=0}^7 (b_i^{ll})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{ll})^2 + 2^8 \sum_{i=8}^{15} (b_i^{ll})^2 \\
& + 2^8 \sum_{i=0}^7 (b_i^{lh})^2 + 2^4 \sum_{i=0}^7 (b_i^{lh})^2 + 2^8 \sum_{i=8}^{15} (b_i^{lh})^2 + 2^4 \sum_{i=8}^{15} (b_i^{lh})^2 \\
& + 2^4 \sum_{i=0}^7 (b_i^{ll})^2 + \sum_{i=0}^7 (b_i^{ll})^2 + 2^4 \sum_{i=8}^{15} (b_i^{ll})^2 + \sum_{i=8}^{15} (b_i^{ll})^2
\end{aligned}$$